

E1 Path dependence

Matched wealth; different histories
Reference domain + portfolio ranking

E2 Finite-sample gradient estimation

Four fixed targets; matched expected cost
Bias / variance / CLT / MSE scaling

E3 Continuous online tracking

Same setup; slow dynamic state path
Residual + DLR, cumulative / average

E4 Reference ablation and robustness

Reference dynamics / horizon / optimizer scale
Where does the tracking advantage strengthen or weaken?

E5 Real-market portfolio evaluation

Time-ordered A-share backtest
Risk-adjusted performance + trading intensity

E6 Investor behavior prediction

Held-out sell / hold opportunities
Held-out prediction + behavioral alignment

E7 Simulated investor interaction

Heterogeneous agents informed by E6
Adoption + satisfaction

